



Course Objective and Outcome Form

Department of Electrical and Computer Engineering

School of Engineering and Physical Sciences

North South University, Bashundhara, Dhaka-1229, Bangladesh

1. **Course Number and Title:** CSE 323 Operating System Design

2. **Number of Credits:** 3

3. **Type:** Core

4. **Prerequisites:** CSE 225, CSE 332

5. **Contact Hours:** 3 hours per week

6. **Instructor:** Farhan Tanvir Khan

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Office Hours: ST 9:40 AM - 2:30 PM, RA 9:40 AM - 11:10 AM

7. **Course Summary:** Operating Systems Design: An introduction to the structure of modern operating systems. Topics include operating systems structure, asynchronism, mutual exclusion, deadlocks, monitors, process state transition, interrupts, context switching, storage management for both real and virtual storage, processor scheduling, multi-processing, auxiliary storage management, computer systems performance, network and security.

8. Course Objectives:

The objectives of this course are:

1. Compare, Analyze and judge various Operating Systems building blocks focusing on process management and memory management.
2. Building a system solution on any hardware platform that is able to talk to hardware in a group project.
3. Demonstrate a contemporary Operating System where students look at Android OS.

9. Course Outcomes (COs):

Upon Successful completion of this course, students will be able to:

Sl.	CO Description	Weight (%)
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1.	solve problems and make judgment about certain constructs of OS components would be used to design a resource constrained system	70
2.	design a system that communicates with virtual hardware components simulated with the QEMU emulator	20
3.	compare various aspects of operating system design using xv6 as the reference platform, focusing on how fundamental OS mechanisms (process management, memory management, file systems, system calls, and scheduling) are implemented in resource-constrained systems	10

10. Mapping of CO-PO:

Sl.	CO Description	Program Outcome	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
CO1	Solve problems and make judgement about certain constructs of OS components would be used to design a resource constrained system	a	Cognitive/Evaluate	Lecture	Quiz, Midterm
CO2	Design a system that communicates with virtual hardware components simulated with the QEMU emulator	c	Cognitive/Create	Project	Demonstrate

CO3	Compare various aspects of operating system design using xv6 as the reference platform, focusing on how fundamental OS mechanisms (process management, memory management, file systems, system calls, and scheduling) are implemented in resource-constrained systems	m	Cognitive/Analyze	Lecture	Presentation/Report
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11. Resources

Text books:

N o	Name of Author (s)	Year of Publicat io n	Title of Book	Editio n	Publishe r' s Name	ISBN
1	Abraham Silberschatz , Peter B. Galvin, Greg Gagne	2018	Operating System Concepts	10 th	Wiley	978-1- 119- 32091-3
2	Andrew S Tanenbaum, Albert S Woodhull	1987	Operating Systems Design and Implementation	3 rd	Prentice- Hall	ISBN- 13:9\ 8- 01314293 83

Reference books:

N o	Name of Author (s)	Year of Publicat io n	Title of Book	Editio n	Publishe r' s Name	ISBN
1	G. Blake Meike	2018	Inside the Android OS: Building, Customizing, Managing and Operating Android System Services (Android Deep Dive)			
2	Kay A. Robins, Steve Robins	2003	UNIX Systems Programming: Communication, Concurrency	2 th	Prentice Hall	ISBN-13: 978- 013442 4071

			and Threads			
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12. Weight Distribution among Assessment Tools:

Theory

Assessment Tools	Weight (%)
Quizzes	20
Assignment	10
Midterm	30
Final Exam	30

Term Project and Presentation	10
Total	100

13. Grading policy: As per NSU grading policy available in

<http://www.northsouth.edu/academic/grading-policy.html>

Students will work in groups of **two starting from the midterm exams to complete a project extending the xv6 operating system on the QEMU emulator. Each group will implement new features or modifications in the kernel or user programs and test them within QEMU. At the end of the project, groups will present their work by demonstrating their changes running in the emulator, explaining their implementation, design decisions, and the OS concepts involved. The focus will be on clear demonstration of functionality, understanding of the underlying OS mechanisms, and the ability to explain how their modifications work “under the hood.”

Project Break Down:

1. Project Proposal (1-3 pages)
2. Project Report (15 pages, Font-Times New Roman, Font Size = 12, Line Spacing = 1)
3. Demonstration